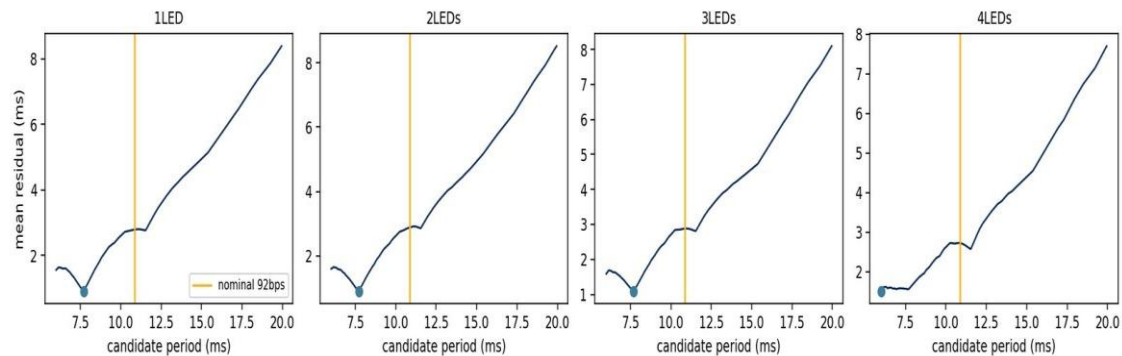


Synchronization Analysis — Candidate Timing

1. TWO INDEPENDENT METHODS, SAME CONCLUSION

Autocorrelation of the denoised signal found no clean periodic peak near the nominal 92 bps bit period (10.87ms) for any of the 4 videos — dominated instead by short-lag smoothness within plateaus.

A second, independent method — searching for the candidate period that best explains observed transition intervals as integer multiples of itself — consistently found a best-fit period of 6-7.7ms, not 10.87ms.

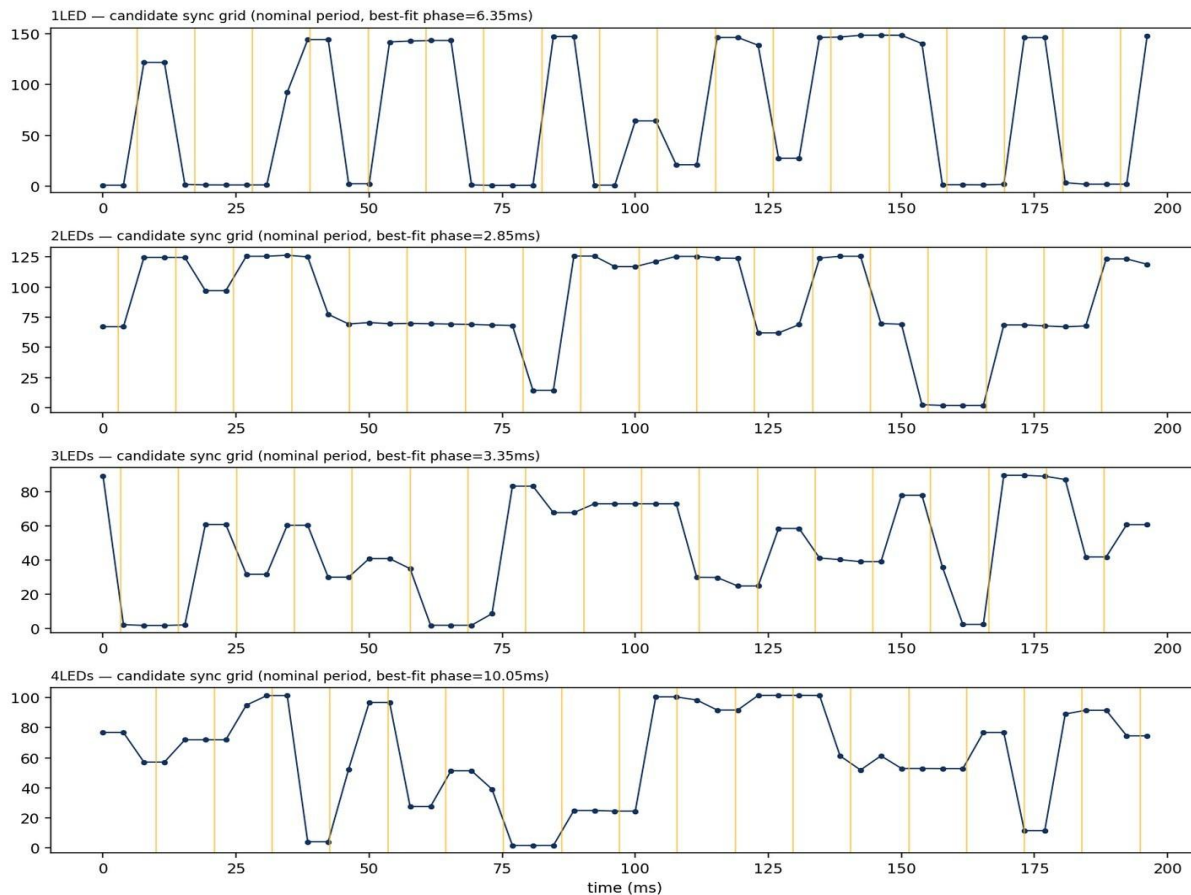


The empirical minimum (blue dot) sits below and to the left of the nominal rate (gold line) in all 4 videos. 7.7ms is suspiciously close to 2x the camera's own frame period (3.846ms), strong evidence the frame grid is being measured, not the transmitter's true clock.

2. FORCING THE NOMINAL RATE: HOW BAD IS THE FIT?

File	Best-Fit Phase	Mean Residual	% of Period
1LED_92bps.mp4	6.35 ms	2.703 ms	24.9%
2LEDs_92bps.mp4	2.85 ms	2.687 ms	24.7%
3LEDs_92bps.mp4	3.35 ms	2.668 ms	24.5%
4LEDs_92bps.mp4	10.05 ms	2.678 ms	24.6%

A ~25% mean residual relative to the period is a poor fit, this is not a usable symbol clock as measured. The visual below makes the same point directly.



The candidate grid (gold lines) does not reliably land on real transitions — several fall mid-plateau. Low confidence, shown honestly rather than cherry-picking a window that looks better.

3. PRIMARY HYPOTHESIS

260 fps gives only 2.826 samples per bit at 92 bps — below the level needed for robust clock recovery from simple per frame-mean thresholding (5-10+ samples/symbol is generally considered safe). This is an under-sampling problem, not a noise problem — Day 10 already ruled out noise as the limiting factor (SNR 57x-318x).

4. ALTERNATE HYPOTHESES TO TEST BEFORE CONCLUDING THIS RATE IS UNRECOVERABLE

- Rolling-shutter row-level extraction — each frame is read out row-by-row over the frame period; individual rows may carry finer time resolution than the per-frame mean. This is the standard technique in the Day 2 literature review and is the most promising next step.
- The nominal 92 bps may not be exact — filenames are often rounded; the true rate could differ enough to shift which period best fits the data.
- Line coding may not be simple NRZ — e.g. Manchester coding doubles the guaranteed transition rate, which would change the expected interval statistics used here.

5. CANDIDATE TIMING CARRIED FORWARD

Nominal period 10.87ms with the per-video best-fit phases in Section 2, usable as a low-confidence starting point for Day 15's baseline decoder, explicitly flagged as such rather than presented as solved. Day 16-17 should prioritize rolling-shutter row-level extraction rather than trying to force this frame-level clock to work harder.